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COLLEGE OF COMPUTER, INFORMATION AND COMMUNICATIONS TECHNOLOGY

Worksheet 2: LOGIC AND CONTROL FLOW

Instructions: Complete the programming problems below by writing Python code in a single file saved as: `lastName_worksheet2.py`.

Separate each problem using clear block comments:

```
# =====  
# PROBLEM 1: Grading System  
# =====
```

Problem 1: Grading System

Write a program that prompts the user for a student's numerical grade (0 to 100, supporting decimals via `float()`).

Use if-elif-else statements with chained comparisons to print:

- Grade ≥ 90 : "Excellent! You are a top student."
- $80 \leq \text{Grade} \leq 89.99$: "Good job! You passed with flying colors."
- $75 \leq \text{Grade} \leq 79.99$: "You passed, but there's room for improvement."
- Grade < 75 : "You failed. Please review the material."

Handle invalid inputs (grades < 0 or > 100) by printing an error message.

Problem 2: Simple Login Check

Write a program that uses logical operators to verify credentials.

1. Hardcode baseline credentials:

`correct_user = "admin"`

`correct_pass = "secret123"`

2. Prompt the user for their username and password.

3. Use a single if statement with the 'and' operator to verify both inputs.

4. If valid, print: "Login successful."

5. Otherwise, print: "Invalid username or password."

Problem 3: Guess the Number

Write a guessing game using a while loop:

- Set secret_number = 7.
- Use an infinite loop (while True:) to repeatedly prompt the user for their guess (int()).
- If the guess matches secret_number:
Print "Correct! You win!" and use the 'break' keyword to exit.
- If incorrect:
Print "Try again!" and allow the loop to repeat.

Problem 4: Calculating Total and Average Score

Given the following list of quiz scores:
scores = [18, 20, 19, 21, 23, 20]

- Initialize total_score = 0.
- Use a 'for' loop to iterate through the list and accumulate the total sum (Do NOT use Python's built-in sum() function).
- Calculate the average using total_score and len(scores).
- Print both values using an f-string formatted to 2 decimal places.

Example Output:
Total Score: 121
Average Score: 20.17

